



Enterprise Crypto Forensics & Real-Time Blockchain Analytics Platform

Smart India Hackathon 2026 • Problem Statement PS-26183
Indian Cyber Crime Coordination Centre (I4C) • Ministry of Home Affairs, Government of India

Platform: CyberTrace Enterprise Engine	Global Benchmarks: Chainalysis, TRM Labs, Arkham
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Repository: github.com/Hidayatulla268/CyberTrace	Live Demo: hidayatulla268.github.io/CyberTrace/

Executive Summary & Vision:
CyberTrace provides a unified, enterprise-grade blockchain analytics platform matching global capabilities (Chainalysis Reactor, TRM Labs, Arkham Intelligence, Elliptic) while solving the operational and legal needs of **Indian Law Enforcement Agencies (I4C, State Cyber Cells, 1930 Helpline)**. From a single suspect wallet address, the system traces multi-hop peeling chains, detects cross-chain bridge hops, de-anonymizes mixers, correlates syndicate nexus networks, and generates Section 91 CrPC freeze orders.

THE CORE FORENSIC WORKFLOW (USP)



1. Enterprise Feature Matrix & Global Comparison

Feature & Capability	Global Benchmark	CyberTrace Implementation
■ Smart Multi-Chain Scanner	Etherscan / Blockchair	Multi-token balances (USDT, ETH, BTC, TRX, INR), ledgers & address tagging.
■ Stolen Money Tranche Tracker	Chainalysis Reactor	Hop-by-hop tranche tracker with retention percentages & CSV export.
■■ Animated Fund-Flow Graph	MetaSleuth / Breadcrumbs	Interactive SVG flow canvas with particle animation, pan/zoom & node popovers.
■ Time-Travel Playback Scrubber	Breadcrumbs / MetaSleuth	Play/pause scrubber animating transactions chronologically (Hour 0 to Hour 24).
■ Cross-Chain Bridge Tracker	TRM Labs Forensics	Tracks hops across Ethereum, Tron (TRC-20), BSC, Bitcoin via Across & FixedFloat.
■■ 100k+ Entity Directory	Arkham Intelligence	Deanonymization database: CEXs, Lazarus Group, Tornado Cash, Darknet markets.
■■ Mixer Demasking Engine	Elliptic Investigator	Tornado.Cash 10/100 ETH deposit-withdrawal matching via gas relay lineages.
■■ OFAC Sanctions Screener	Chainalysis KYT	Real-time compliance checks against US OFAC SDN, EU Sanctions & FIU-IND alerts.
■ Fraud DNA™ Syndicate Matcher	Proprietary Innovation	Detects zero-day unreported wallets via 8-D behavioral sequence vector matching.
■■ Live Mempool Watchlist	SlowMist MistTrack	Live mempool ingestion stream with visual pulses and automated threat flags.
■ Case Manager & Evidence Vault	Chainalysis Case Manager	Multi-case workspace: assign police officers, add encrypted case notes & export.
■ Subpoena Dispatcher	I4C Law Enforcement	1-click Section 91 CrPC, US CLOUD Act, and MLAT freeze orders with API dispatch.

2. Deep-Dive: Fraud DNA™ & Zero-Knowledge Demasking

Fraud DNA™ (Zero-Day Detection):

Traditional tools only flag known addresses. Fraud DNA™ computes an 8-dimensional behavioral vector (Transfer Cadence, Peeling Ratio, Hop Depth, Tranche Size, CEX Gateway, Gas Sponsorship). When an unreported wallet appears, CyberTrace detects: **■ Potential Unreported Fraud Wallet Detected — 91% match with Campaign #CYB-2048'.**

Mixer & Obfuscation Demasking:

De-anonymizes privacy pools like Tornado.Cash by identifying common relayer gas dispatchers (e.g. 0xRelay99B) and correlating leaf index deposit timings against withdrawal blocks, reducing the anonymity set from 10,000 to single-digit candidate targets.

3. Courtroom Admissibility & Law Enforcement Integration

Section 91 / 102 CrPC Legal Framework:

Under Indian Criminal Law, freezing cryptocurrency on centralized exchanges requires formal legal notices issued by Police Officers or Magistrates. CyberTrace automates this workflow by pre-populating deposit transaction hashes, internal memo UUIDs, and estimated loss figures directly into Section 91 CrPC notice templates. All exported dossiers include SHA-256 cryptographic timestamps ensuring chain-of-custody compliance under the Indian Evidence Act.

4. Access & Repository Resources

GitHub Repository:	https://github.com/Hidayatulla268/CyberTrace
Live Website (GitHub Pages):	https://hidayatulla268.github.io/CyberTrace/
Documentation PDF:	CyberTrace_Platform_Guide.pdf